For this assignment, the original program reads a healthcare dataset, checks for invalid information, removes duplicate records, creates summary statistics, replaces patient names with identification numbers, and encrypts the cleaned data. I used PyCharm to run the program, review its functions, and made three improvements. The goal was to make the program more accurate and reliable.

The first improvement I made was changing when the program removes duplicate records. Originally the script removed duplicates before standardizing certain text values. I moved the duplicate removal code so that it runs after the program formats categories such as gender, medical condition, and insurance provider. This helps the program recognize duplicate records even when the category values have different capitalization. While doing this I ran into an indentation problem that caused the duplicate removal code to run too many times inside a loop. After fixing the indentation the program correctly reported that it removed 534 duplicate records.

My second improvement was adding better error handling when the program reads the CSV file. I used try and except statements to check for problems such as a missing file, an empty file, or a file that can’t be read properly. Those statements provided clearer error messages so that someone using the program can understand what went wrong. This is helpful because healthcare data programs should be able to identify problems with their input files instead of continuing with incorrect information.

The third improvement was strengthening the age validation process. The original script checked whether patient ages were between 0 and 120, but it also rounded decimal ages to whole numbers. For example, an age of 25.7 could become 26. I changed the code to recognize decimal ages as invalid and replace them with missing values instead of rounding them. This helps prevent the program from changing questionable information into a value that appears valid.

After making those improvements, I ran the program again in PyCharm. The program read 55,500 records, and retained 54,966 records, and removed 534 duplicates. The final script ran without any errors.